



Long-distance contacts and social networks of the Upper Palaeolithic humans in the North-Western Caucasus (on data from Mezmaiskaya Cave, Russia)

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ABSTRACT

The article is addressed to the issue of long-distance contacts and social networks between the Upper Palaeolithic of the North-western Caucasus and other regions, basing on finds (artifacts from mammoth tusk, azurite sandstone pebble, Black Sea marine mollusc shells, and obsidian artifacts) from the Upper Palaeolithic layers at Mezmaiskaya Cave. Personal decorations from mammoth tusk found at Mezmaiskaya suggest rare contacts between culture-unrelated Upper Palaeolithic groups in the North-western Caucasus and Russian plain. Recurrent obsidian procurement from two specific remote obsidian sources at Mezmaiskaya suggests that there were two well-established social networks between Upper Palaeolithic humans in the North-western Caucasus and related human groups in the North-central and South Caucasus. The evidence of obsidian transport from Mezmaiskaya outlines borders of the Upper Palaeolithic cultural area with similar human population and culture in the Caucasus, within which the Upper Palaeolithic groups in the North and South Caucasus had regular social-cultural relations, including exchange of obsidian raw material. Finds of Black Sea marine shells and an azurite sandstone pebble at Mezmaiskaya are discussed as non-utilitarian objects derived from remote areas located on the paths of the two obsidian exchange networks that likely outline directions and limits of direct movements of Upper Palaeolithic humans from the North-western Caucasus.

1. Introduction

Numerous studies show that important social-demographic traits found in all Upper Palaeolithic (UP) regions of Europe are increasing population density and the concomitant size of residential groups, and changes in mobility strategies of UP groups based on evidence of the transportation of raw materials over several hundred km (e.g., [Gamble, 1986](#); [Bocquet-Appel et al., 2005](#); [Féblot-Augustins, 2009](#)). Many researchers consider the long-distance transportation of raw materials and various non-utilitarian objects as evidence for the formation of culturally proscribed areas and organization of complex social networks that provided intergroup cultural transmission, trade, and exchange (e.g., [Gamble, 1999](#); [Mills et al., 2013](#); [Horiuchi and Takakura, 2019](#)). Palaeogenomic studies also reveal various patterns of genetic (mating)

interactions between different UP groups in Europe and Asia ([Seguin-Orlando et al., 2014](#); [Fu et al., 2016](#); [Sikora et al., 2017](#)), suggesting that the social organization of UP humans resembled those of living hunter-gatherers, with limited relationships within residential groups embedded in large-scale social networks that were associated with intergroup cultural contacts and mating. Generally, advances in the study of the UP society suggest that the complex social structures that interweave marriage relations and access to natural resources via social networks, and which characterize all modern human societies, emerged after 30 ka ([Foley and Gamble, 2009](#)).

The occurrence of rare lithic artifacts made from exotic rocks, i.e. brought from distance over 100–200 km to the site, is a highly informative source concerning long-distance human mobility, exchange and other social contacts between UP human groups ([Féblot-Augustins,](#)

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2009; Kuhn, 2004; Aubry et al., 2012). Also, unusual, so called 'non-utilitarian' (i.e., not related to household or life sustaining economic activities) can have significance as evidence of long-distance contacts between UP groups. Many examples of unusual lithic artifacts are known in the UP of West and Central Europe. During the earlier UP unusual materials collected mostly within local environments, but in the later UP the high value of some exceptional materials led to increasingly distant procurement, likely involving long-distance networking (Moncel et al., 2012). On the central Russian Plain, nonlocal materials including lithics, marine shells, and amber, which are found in many UP sites, were obtained likely through specialised procurement in some areas and moved through exchange via long-distance trade networks across the whole territory of Russian Plain (Soffer, 2013).

Modern palaeogenomic (Jones et al., 2015; Fu et al., 2016) and archaeological (Golovanova and Doronichev, 2020) data from the Caucasus show that the population history and cultural development of

the UP humans in the region are specific and differ from those in Europe and West Asia in several respects. Comparisons of the UP assemblages from the North-western (NW) and South-western (SW) Caucasus reveal that the UP industries on both slopes of the Greater Caucasus demonstrate a substantial similarity, indicating dispersal of close UP groups over the whole western Caucasus region. At present, there is also a wide consensus among researchers working directly in the Caucasus that the UP industry of this region does not represent a variant of the Aurignacian or Gravettian industries (e.g., Meshveliani et al., 2004; Bar-Yosef et al., 2006, 2011; Adler et al., 2008; Golovanova and Doronichev, 2012, 2020; Pleurdeau et al., 2016; Kandel et al., 2017).

In this paper we discuss several kinds of archaeological data derived from previous and ongoing research at Mezmaiskaya Cave, in the NW Caucasus, that provide increased knowledge about long-distance contacts and social networks that existed before the Last Glacial Maximum (LGM, ~25–20 ka cal BP) between the UP humans in this



Fig. 1. Map showing the location of Mezmaiskaya Cave in the NW Caucasus relative to UP sites in the South Caucasus, Biryuchiya Balka-2 on the southern Russian Plain, and identified lithic raw material sources. Legend: 1 – cave sites, 2 – Biryuchiya Balka-2 open-air site, 3 – obsidian sources: A – Chikiani; B – Zayukovo (Baksan), 4 – azurite sources: C – Khudes deposit; D – Tyrnyauz deposit. UP cave sites in the Caucasus: 1–2 – Mezmaiskaya and Korotkaya caves; 3 – Akhshtyrskaya cave; 4 – Apiancha cave; 5–6 – Dzudzuana cave and Ortvale klde rockshelter; 7 – Bondi cave; 8–9 – Satsurbliia and Samertskhle klde caves.

region and neighboring regions in the North-central (NC) and SW Caucasus, and the southern Russian Plain.

2. Research background and methods

2.1. Background data

Mezmaiskaya Cave located 1310 m asl, in the NW Caucasus, Russia, is a reference Middle Palaeolithic (MP) and UP site in the Caucasus (Fig. 1). Since 1987, when L. Golovanova started excavations at the site, about 100 m² have been excavated (see SI; Fig. S1) to the depth of 5 m. So far, 6 Holocene and 20 Pleistocene strata have been identified over the excavation area, including seven MP levels, six UP (1A-1, 1A-1/1A-2, 1A-2, 1B-1, 1B-2, and 1C, from top to bottom) and two Epipalaeolithic levels 1–3 and 1–4 (Fig. S2). The total series of 25 radiocarbon, mostly accelerator mass spectrometry (AMS) dates, applying the IntCal20 dataset, defines with a 68.3% probability the calendric age of the Layer 1C dating from 39.9 to 36.4 ka cal BP to 33.4–31.2 ka cal BP; Layer 1B – 31.1–27.5 ka cal BP; Layer 1A-2 – 27.3 ka cal BP; Layer 1A-1/1A-2 – 24.7 ka cal BP; and Layer 1A-1 – 25.4–23.2 ka cal BP, or to the onset of the LGM in the Caucasus (Table S1).

In this paper we report on some archaeological materials (artifacts from mammoth tusk, azurite sandstone pebble, Black Sea marine mollusc shells, and obsidian artifacts) recovered from the UP sequence at Mezmaiskaya Cave. We discuss the results of previous and new analyses of these materials as a basement for the identification and interpretation of long-distance contacts and social networks that existed between the Early Upper Palaeolithic (EUP; ~40–30 ka cal BP) and Late Upper Palaeolithic (LUP; ~30–24/23 ka cal BP) human groups in the NW Caucasus and other regions in the Caucasus and beyond.

2.2. Methods

The numbers of obsidian artifacts and their distribution in various UP layers at Mezmaiskaya Cave are reported in Supporting Information and shown in Tables S3–S7. The obsidian artifacts from Mezmaiskaya were analyzed using the x-ray fluorescence (XRF) method initially at the Archaeological XRF Laboratory, University of California, Berkeley. The bulk of the samples from Mezmaiskaya were analyzed at the Geoarchaeological XRF Laboratory in Albuquerque, New Mexico, USA, by M.S. Shackley, both analyses using the ThermoScientific *Quant'X* EDXRF instruments. Instrumental methods are available in (Shackley, 2005) and here: <http://swxrlab.net/anlysis.htm>.

The pebble of azurite-quartz sandstone found at Mezmaiskaya was studied at Russian Geological Research Institute (VSEGEI), St. Petersburg, Russia, by V.F. Sapega and N.P. Vinogradova. The following methods and equipment were applied to study the sample: a) thin-section petrography, b) x-ray diffraction (XRD) analysis, and c) electron microprobe (EMP) analysis. XRD analysis was performed using x-ray diffractometer DRON-6, equipped with an x-ray tube with a cobalt anode, under the analytical condition: wavelength $\lambda = 1.79021$ Å, voltage 35 kV, and current 25 mA. Processing of recorded spectra and identification of phases were performed, using the PDWin-4 software and the JCPDS International database. Peaks were determined using the Rietveld method. EMP analysis was performed from a polished thin section using CamScan MX-2300 microprobe with Link AN-10/85S adapter ($K\alpha$ radiation, accelerating voltage 20 kV, probe diameter 1×1 µm, exposure time 70 s; sensitivity 0.03–0.04 wt%).

The species definitions for marine mollusc shells from Mezmaiskaya were made at the Institute of Geography, Moscow, Russia, by A.L. Chepalyga. The identification of artifacts made from mammoth tusk was accomplished using a standard criterium (e.g., Espinoza and Mann, 1999), such as the texture (inner lines known as the Schreger lines that are easily seen inside the artifacts from mammoth tusk). Detail identifications of all artifacts and marine mollusc shells from Mezmaiskaya Cave that are discussed in the text are provided in Tables S2.

3. Analytical materials and results

3.1. Artifacts made of mammoth tusk

Artifacts made from mammoth tusk that are found in Mezmaiskaya Cave are of particular interest, because no evidence of mammoth hunting is known in the UP sites in the NW Caucasus and no artifacts from mammoth tusk are found in other UP sites in the region. At Mezmaiskaya, artifacts made from mammoth tusk were found only in the UP layers 1B, 1A-2 and 1A-1/1A-2.

In layers 1B and 1A-2, there are three small ($10 \times 8 \times 2$; $7 \times 10 \times 2$, and $9 \times 7 \times 2$ mm) flat stripe-beads made of mammoth tusk. They have a roughly square shape, with three rounded sides and one straight side, and holes with similar diameter (3, 4 and 2 mm) perforated from one or two sides (Fig. 2 – 2–4). In Layer 1A-2, there were also found two fragments of points made of mammoth tusk: a fragment of large ($74 \times 13 \times 3$ mm) point with a flat cross-section, which split along the tool according to the tusk texture (Fig. 2 – 1), and a medium fragment ($53 \times 14 \times 8$ mm) of point with a rounded cross-section.

In Layer 1A-1/1A-2, a bead or stripe ($8 \times 5 \times 2$ mm) made of mammoth tusk was found. It has a sub-rectangular shape with rounded ends, a cross cut located closer to one end where a hole is drilled, and traces of ochre on the surface (Figs. 2 – 6). The bead has close analogies among bone artifacts from Sungir site on Russian Plain (Bader, 1978: p. 168, fig. 113 – 12).

Another bead ($14 \times 5 \times 4$ mm) from Layer 1A-1/1A-2, made of a thin tubular bird bone, with ends having traces of incisions (Fig. 2 – 7), also has analogies in Sungir (Bader, 1978: 168, fig. 113 – 31–35). In addition, a fragment of bone stripe-bead ($8 \times 4 \times 2$ mm) was found in this layer (Fig. 2 – 5). It has an ellipsoid shape, a bi-convex profile, and a hole drilled from one side. Analogies of this find are unknown.

3.2. Rock definition and source identification of azurite sandstone

A unique find for the Caucasian UP, having no analogues in other UP sites in this region, is a small ($35 \times 24 \times 23$ mm) pebble of blue rock (Fig. 3 – 1) found in EUP Layer 1C at Mezmaiskaya Cave and reported for the first time in this paper. Several analyses were made for the sample cut off from this pebble to determine its rock type and probable source of origin.

Based on thin-section petrography of the sample at low magnification, the rock was defined preliminary as quartz sandstone composed mainly of poorly sorted, angular clastic material of different psammitic dimensions, represented by quartz (93%) and green malachite (5%), with admixture of tufogenic lithoclasts, chalcedony, and microfelsite cemented by turquoise of ultramarine color, and with numerous inclusions of iron oxides sporadically staining the turquoise cement in rusty tones (Fig. 3 – 3a, b).

XRD analysis showed that the mineral composition of the sample is composed predominantly of azurite ($49 \pm 5\%$) and quartz ($45 \pm 5\%$), and the rock is quartz-rich azurite sandstone or azurite-quartz sandstone (Fig. 3 – 4; Table 1). A detailed chemical composition of this rock was determined by EMP analysis of five microprobes that had been executed on the clasts and on the matrix in different parts from the thin section of the specimen (Spectra 3–7; Fig. 3 – 2). The spectrum 2 and especially the microprobe spectrum 4 show the most complete chemical composition of the azurite-quartz sandstone pebble from Mezmaiskaya (Table 2).

Azurite ($\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$) is known as a secondary mineral originating from the oxidation zone in several copper deposits associated with carbonate rocks in the North Caucasus (see <https://webmineral.ru>). The Belorechensk deposit in the Belaya River valley, near the former town of Nickel in Adygeya Republic, is the nearest to Mezmaiskaya Cave azurite source, but azurite occurs here only in the form of small crystals. The Tyrnyauz deposit is located 2000–3200 m asl in the Baksan River valley, approximately 40 km east of Mt. Elbrus. Azurite occurs here as a secondary mineral and some pieces of azurite-rich rocks



Fig. 2. Mezmaiskaya Cave. Artifacts from mammoth tusk and bone from UP layers. See Tables S2 for detail provenance of the artifacts. Scale bar = 1 mm.

reach up to $70 \times 80 \times 65$ mm in size. The Khudes deposit is located 2800–3400 m asl in the upper reaches of the Tohan River, at the northern foot of Mt. Elbrus. Azurite occurs here as a secondary mineral, while the chemical composition of ores from the Khudes deposit includes cuprum, argentum and sulfur, and is similar to the chemical composition of azurite from Mezmaiskaya.

This evidence suggests that the Khudes or Tyrnyauz deposits in the Elbrus region are the most likely sources of origin for the Mezmaiskaya's pebble of azurite-quartz sandstone. However, no UP sites are known at elevations above 1700 m asl in the North Caucasus (see Golovanova and Doronichev, 2020), so it is most likely that UP humans could have taken this pebble from secondary alluvial deposits in lower elevations.

3.3. Marine mollusc shells

In the UP layers at Mezmaiskaya Cave, one of the types of finds that were brought to the site from remote areas are marine mollusc shells. According to the definition by A. Chepalyga, they belong to *Trophon muricatus* (Mtd.) or *Nassarius reticulatus* (Linne) and *Cyclope neritea* (Linne). These marine molluscs are commonly known in the Black Sea basin. Some of the shells were previously published (see Golovanova and Doronichev, 2012), but most come from the more recent excavations in the cave and are published here for the first time (Fig. 4 – 4–9).

Small (7–13 mm in diameter) marine gastropod shells of *Trophon muricatus* or *Nassarius reticulatus* without holes, some of which are fragmented or have bad preservation, were found in layers 1C, 1B, 1A-2, and 1A-1 (Fig. 4). Also, the larger marine shells of *Trophon muricatus* or *Nassarius reticulatus* were found, some of which have a pierced hole and were probably used as pendants or beads (Fig. 4 – 1,3). These pierced shells were found only in the UP layers 1B, 1A-2 and 1A-1. Additionally, a pendant or bead made from a *Cyclope neritea* (Linne) shell was found in Layer 1A-2 (Fig. 4 – 2).

3.4. Obsidian artifacts

At Mezmaiskaya, 821 obsidian artifacts (1.9% of the total lithics) were found in the UP layers 1C, 1B, 1A-2, 1A-1/1A-2, and 1A-1 in all excavations (Table 3). Of them, only 317 artifacts were analysed previously (Doronicheva and Shackley, 2014; Doronicheva et al., 2019), and the rest come from the more recent excavations in the cave and are reported for the first time herein. The percent of obsidian artifacts is low in all UP layers, indicating that UP humans in the NW Caucasus rarely

used this exotic raw material, which sources are unknown in this area.

In the EUP Layer 1C, the obsidian artifacts were all individual finds (52 pieces; 0.5% of the analysed assemblage). The absence of cores, core trimming elements (CTEs), and flakes with cortex, along with the prevalence of bladelets/microbladelets, chips/microchips, and retouched tools suggest that obsidian was apparently transported to the cave as ready-to-use laminar blanks or retouched tools. Tools include backed bladelets and one Gravette point.

In Layer 1B, the number of obsidian artifacts increased slightly, although obsidian artifacts were also rare finds (114 pieces; 1.8% of the assemblage). The obsidian artifacts include CTEs and a flake with cortex, suggesting that a limited flaking of obsidian took place in the cave. The tools (15 pieces) include backed bladelets, retouched bladelets and flakes, points on bladelets (Fig. 5 – 6), and one end-scraper on blade (Fig. 5 – 7).

Most of the obsidian artifacts at Mezmaiskaya come from Layer 1A-2 (475 pieces; 2.4% of the total lithic assemblage). The obsidian artifact assemblage includes core (Fig. 5 – 12), CTEs, cortical flakes, flakes (Fig. 5 – 1) and numerous fragments and chips, suggesting that a limited on-site flaking of obsidian took place. Laminar blanks are dominated by bladelets (Fig. 5 – 3, 10) and microbladelets (Fig. 5 – 2). The retouched tools from obsidian are characterized by a predominance of backed and retouched bladelets and microbladelets (Fig. 5 – 11). Also, a Gravette point (Fig. 5 – 5) and burins made on obsidian blanks were found.

In Layer 1A-1/1A-2, the percentage of obsidian artifacts increased slightly (3.0%; 121 pieces). The obsidian artifacts include CTEs, cortical flake, blade (Fig. 5 – 8), bladelettes/microbladelets, small fragments, and chips, suggesting a limited on-site flaking of obsidian. The retouched tools are rare and include only backed bladelets and one burin.

Layer 1A-1 yielded fewer obsidian artifacts (59 pieces; 2.0%). The obsidian artifact assemblage is dominated by blades, bladelets (Fig. 5 – 9), microbladelets, chips/microchips, and small fragments. The rare retouched tools include retouched microbladelets, one retouched bladelet (Fig. 5 – 4), and one Gravette point.

The sourcing studies of 30 obsidian artifacts from UP layers at Mezmaiskaya Cave (Doronicheva et al., 2013, 2019; Doronicheva and Shackley, 2014) identified two obsidian sources (Fig. 1), one (Zayukovo or Baksan) located in the NC Caucasus, about 80 km NE from Mt. Elbrus (~250 km linearly SE from Mezmaiskaya) and another (Chikiani or Chikiani-Paravani) located on the Javakheti Plateau in southern Georgia, in the northern Lesser Caucasus (~450 km linearly SSE from

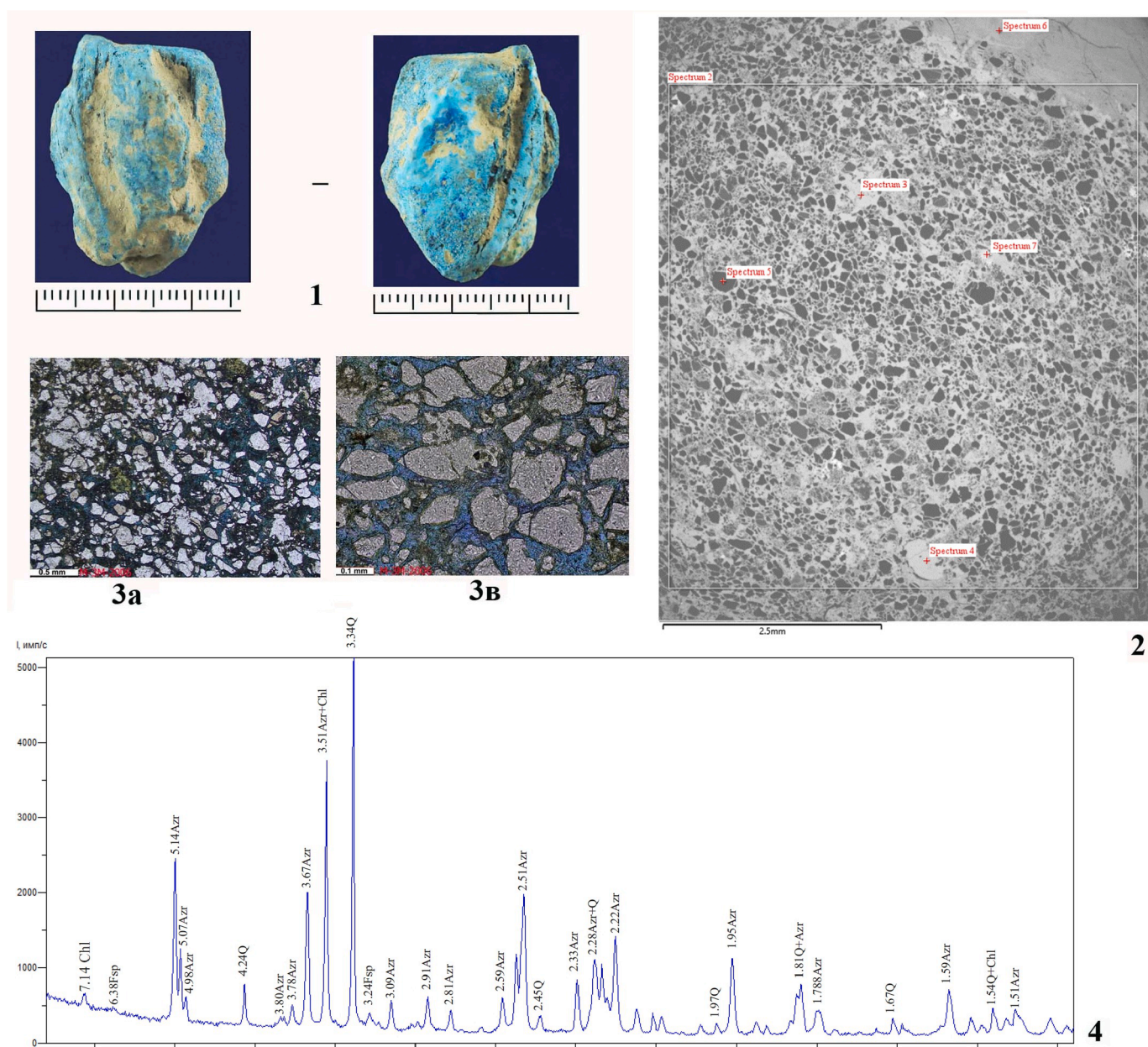


Fig. 3. Mezmaiskaya Cave. XRD and EMP analytical results for the pebble of azurite-quartz sandstone. See [Tables S2](#) for detail provenance of the pebble. 1 – photo of blue rock pebble from Layer 1C; 2 – photo of the thin section specifying the five spot positions (spectra 3–7) where the analyses by EMP were executed; 3a, b – color micrographs from XRD analysis, showing a petrographic texture of the rock at different magnification; 4 – diffractogram showing spectra of chemical elements ([Table 1](#)) identified by XRD analysis in the rock: Azr – azurite, Chl – chlorite, Fsp – feldspar, Q – quartz. The list of minerals, identified by EMP analysis of the rock, is shown in [Table 2](#). See [Tables S2](#) for detail identification of the pebble. Scale bar is 1 mm (1), 2.5 mm (2), 0.5 mm (3a), and 0.1 mm (3b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Mineral composition (wt%) of azurite-quartz sandstone from Mezmaiskaya, identified by XRD.

Mineral	Composition (wt%)
Azurite	49 ± 5
Quartz	45 ± 5
Feldspar	3 ± 1
Chlorite	~2

Mezmaiskaya). For this study, we analysed additional 25 artifacts from UP layers at Mezmaiskaya Cave, of which 18 were assigned to Zayukovo and 7 to Chikiani 3a ([Tables S8](#)). The new results confirm the previous

conclusion that UP humans in the cave used exclusively only the two obsidian sources— Zayukovo (27 samples analysed in total) and Chikiani (28 samples analysed in total)—identified earlier ([Figs. 6 and 7](#)).

We use Ba/Rb and Ba/Zr plots here specifically to compare to the biplots used by [Biagi et al. \(2017\)](#) and [Biagi and Gratuze \(2016\)](#) since their study of the Chikiani source is the most extensive and recent study of that source. These elements are useful in studies of volcanic rocks, particularly rhyolite (obsidian) since they are incompatible elements (light-ion lithophile elements) with high or large ionic charges that have difficulty finding mineral hosts and thus remain in the melt for a considerable period of time, and at eruption are a direct image of the melt composition (see [Glascok, 2011:172](#) for a discussion of these elements in XRF analysis).

Table 2
Chemical composition of azurite-quartz sandstone from Mezmaiskaya, identified by EMP analysis.

Chemical elements	Spectrum 2		Microprobe spectrum 4	
	Weight %	Atomic %	Weight %	Atomic %
cuprum (Cu)	27.63	13.81	33.04	22.01
oxygen (O)	29.98	59.53	18.25	48.28
carbone (C)			6.85	24.14
silicium (Si)	19.82	22.42	0.42	0.63
aluminium (Al)	2.38	2.81	0.24	0.37
kalium (K)	1.38	1.12	0.05	0.06
sulfur (S)	0.16	0.16	3.21	4.24
ferrum (Fe)	0.25	0.14	0.08	0.06
argentum (Ag)			0.19	0.08
phosphorus (P)			0.05	0.07
calcium (Ca)			0.06	0.06
Total	81.61	100	62.44	100

The data now available show that Zayukovo and Chikiani are the only two obsidian sources that were used exclusively by humans in Mezmaiskaya Cave during the entire UP. Also, the preliminary results show that the ratio of Zayukovo/Chikiani obsidian varied during the UP sequence at Mezmaiskaya: from 6:4 in layer 1C, 4:7 in layer 1B, to 17:17 in layer 1A (including all sub-layers), based on the small number of samples analysed. The sourcing studies have been conducted for only 6.8% of the total obsidian artifacts from UP layers at Mezmaiskaya. We plan to continue the research using a larger sample to better evaluate the

significance and possible changes in exploitation of the Zayukovo and Chikiani obsidian in the cave.

4. Discussion

4.1. Significance of UP obsidian transport/exchange networks

Modern research indicates distinct regional peculiarities and a specific pathway of UP development in the Caucasus, represented by the 'Caucasus Upper Palaeolithic industry' (Golovanova and Doronichev, 2020). In both the North and South Caucasus, UP assemblages are characterized by a highly developed microblade (bladelet) technology aimed at the production of numerous bladelets and microbladelets (width <12 mm), and, to a lesser degree, larger blades. In many UP sites, prismatic unipolar cores prevail, although narrow-fronted cores are also quite abundant in some sites. The Caucasus UP industry is also characterised by the prevalence of tools made on bladelets, including mostly various retouched and backed bladelets, and typical Gravette and microgravette straight-backed points made by blunted retouch, as well as needle-like retouched and symmetrical backed points. However, typical Dufour bladelets have not been found in securely excavated and dated stratified UP sites in this region. End-scrapers and burins are represented, but are much less abundant than backed and retouched bladelets, and end-scrapers outnumber burins in most Caucasian UP assemblages. The Caucasus UP industry also shows a wide assortment of bone tools, including characteristic double-pointed bone points with



Fig. 4. Mezmaiskaya Cave. Marine mollusc shells from UP layers 1C – 1A-1. See Tables S2 for detail provenance of the shells. Scale bar = 1 mm.

Table 3

Mezmaiskaya Cave. Assemblage composition of obsidian artifacts from UP layers, 1997–2019 excavations.

Layer	Cores	Small fragments	CTE	Flakes			Laminar blanks			Chips/microchips	Tools	Total obsidian artifacts	Percent in the total excavated assemblage n (%)
				Primary (50–100% cortex)	Cortical (1–50% cortex)	Without cortex	Blades	Bladelets	Microbladelets				
1A-1	–	8	–	–	1	6	4	8	2	16/10	4	59	2873 (2.0)
1A-1/ 1A-2	–	35	2	1	–	4	7	13	12	22/19	6	121	3984 (3.0)
1A-2	1	63	7	2	5	22	5	48	31	125/143	23	475	19,528 (2.4)
1B	–	13	2	–	1	5	1	8	9	38/22	15	114	6295 (1.8)
1C	–	3	–	–	–	3	–	3	12	23/4	4	52	9641 (0.5)
Total	1	122	11	3	7	40	17	80	66	224/198	52	821	42,321 (1.9)

**Fig. 5.** Mezmaiskaya Cave. Obsidian artifacts from UP layers. See [Tables S2](#) for detail provenance of the artifacts. Scale bar = 1 mm.

rounded cross-sections, awls, smoothers, and needles. Also, it is characterized by a few common types of personal ornaments, such as pendants or beads made from caprid teeth, stone, or shell, particularly cockleshells and marine gastropod shells.

The LUP assemblages of the Caucasus indicate notable changes in the lithic assemblages and bone artefacts after c. 30 ka cal BP, including the production of more numerous blades in some LUP sites, or the production of bladelets/microbladelets from carinated cores in others. We also see higher variability among stone points, with the addition of Font-Yves and Sakajia points in some sites; a higher diversity of personal ornaments made from bone, mammoth tusk and stone; and the first appearance of bone artefacts with fine geometric ornamentation.

The Caucasian UP assemblages also demonstrate similar strategies of raw material procurement. The results obtained in Mezmaiskaya and comparative data from other UP sites in the Caucasus indicate that (1) the typical area of resource exploitation and mobility of UP groups did not exceed linear distances of about 100 km from the site, and (2) UP humans in the NW Caucasus preferred to exploit high-quality raw materials (varieties of flint) derived from sources located at linear distances of about 30–40 km ([Doronicheva et al., 2013, 2019](#)).

Similar results have been reported for UP obsidian exploitation in Aghitu-3 cave in the Lesser Caucasus—a region where obsidian sources are abundant. The UP humans at Aghitu-3 exploited almost exclusively varieties of obsidian from local sources located at linear distances of about 30–40 km from the site ([Kandel et al., 2017](#)).

Also, as reported above, rare obsidian artifacts (from 0.5% to 3.0% of the total lithics in various layers) were found in all UP layers at Mezmaiskaya Cave. Studies of obsidian and flint artifacts from Mezmaiskaya

suggest a deliberate selection by UP humans of high-quality raw materials transported from distant sources, mainly for the production of bladelets and microbladelets from which to make retouched tools, or to use these blanks as unretouched tools ([Doronicheva and Shackley, 2014; Doronicheva et al., 2013, 2019](#)). The UP humans at Mezmaiskaya used obsidian as an exotic high-quality lithic raw material, which entered the site from great distances and mainly in the form of ready-to-use small laminar blanks (bladelets and microbladelets) or as tools made on bladelets and microbladelets, although in rare cases obsidian was brought as prepared cores.

Research of obsidian artifacts from Mezmaiskaya cave identified two extensive obsidian transportation networks that provided supplies of obsidian to UP humans in the NW Caucasus from two obsidian sources located in the NC Caucasus and South Caucasus ([Fig. 1](#)). The UP humans in the NW Caucasus more or less regularly use obsidian derived from these two obsidian sources during the entire UP, from the initial appearance of AMHs in this area c. 40–36 ka cal BP through the onset of the LGM at c. 25–24 ka cal BP.

Obsidian artifacts are also present but are very rare in some UP sites in the SW Caucasus—the area including west Georgia, Abkhazia and the Sochi region in Russia, and lying on the route from the Chikiani obsidian source in the Lesser Caucasus to Mezmaiskaya Cave in the NW Caucasus. In the SW Caucasus, obsidian artifacts are best studied in west Georgian UP sites, including Dzudzuana cave, where they comprise 0.8–2.3% of the total lithic artifacts ([Bar-Yosef et al., 2011](#)), Ortvale Klde rockshelter, and Satsurblia and Bondi caves ([Adler et al., 2008; Bourdonnec Le et al., 2012; Pinhasi et al., 2014; Pleurdeau et al., 2016](#)), where they comprise ~1.0% of the total lithics. Like Mezmaiskaya, obsidian arrived to the UP sites in the SW Caucasus mostly in the form of small semi-finished products (mainly unretouched bladelets and small blades) and rarer as retouched tools, with only rare evidence for on-site knapping or shaping of tools from obsidian.

Sourcing analyses of several obsidian artifacts ([Bourdonnec Le et al., 2012; Pleurdeau et al., 2016](#)) indicate that most obsidian artifacts analysed from the UP layers in Bondi and Ortvale klde were brought to the sites from the Chikiani obsidian source (about 110–120 km linearly SSE of the sites), the same source used by UP humans in Mezmaiskaya Cave. However, a few samples from the Georgian sites likely originate from other sources located at linear distances of about 200 km to the south (Kars and Sarikamis in eastern Turkey, in Anatolia) and about 150 km SE (Ashotsk or Aghvorik in northern Armenia, in the Lesser Caucasus). The results suggest that UP humans in the SW Caucasus had contacts with areas located at linear distances of >100 km to the south and SE, in the northern Lesser Caucasus and eastern Anatolia.

In Aghitu-3 cave, in the Lesser Caucasus, the prevailing raw material, comprising 69–100% of the overall lithic assemblage in various layers ([Kandel et al., 2017: tab. 4](#)), was obsidian. An XRD study of 1121 obsidian artefacts from Aghitu-3 showed that UP humans procured obsidian almost exclusively (92–99% of the total obsidian artefacts in all UP layers) from two regional volcanoes located at linear distances of about 30–40 km to the north of the site. Rare obsidian artefacts were also

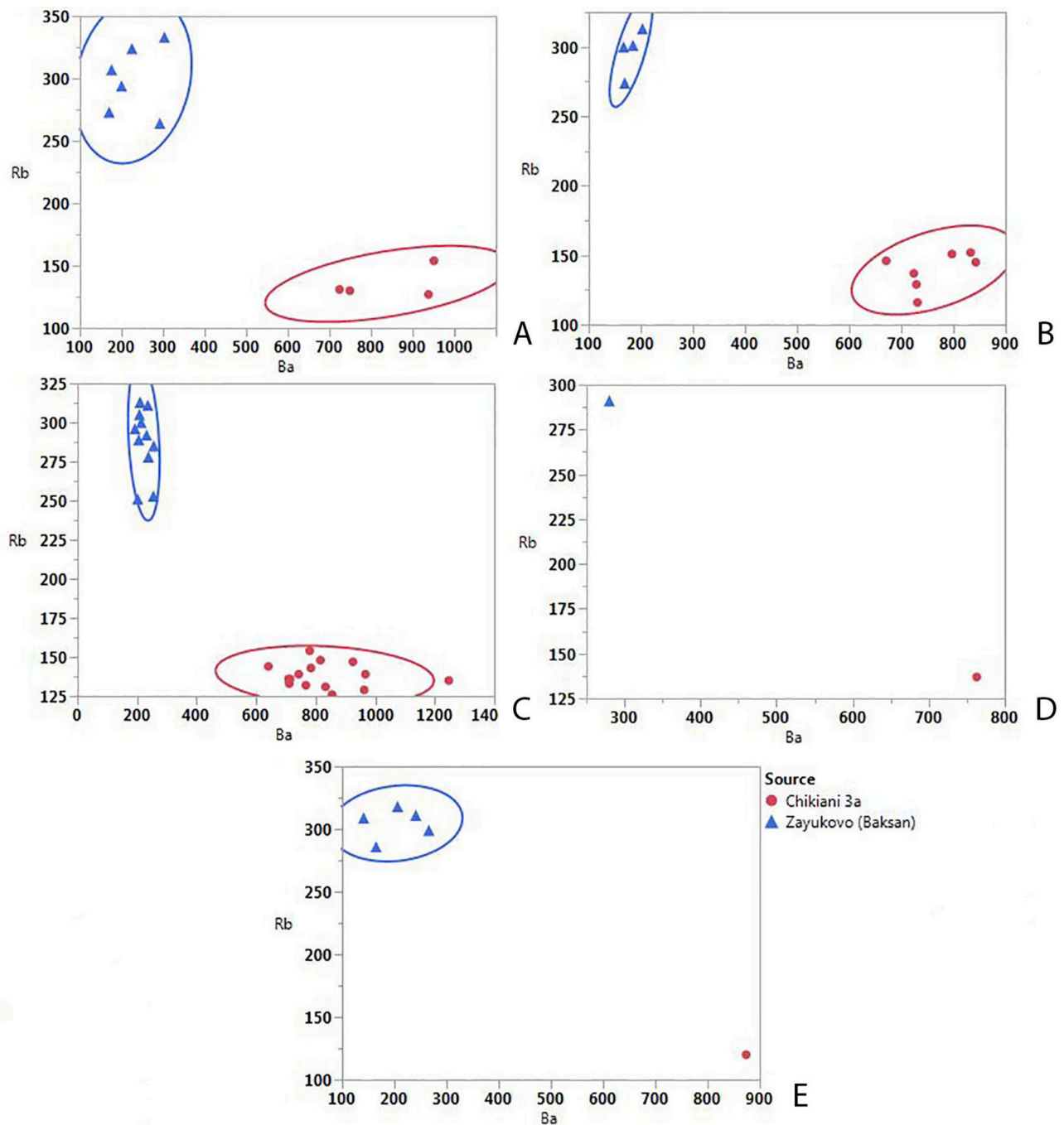


Fig. 6. Ba/Rb bivariate plots of the archaeological obsidian samples from UP layers at Mezmaiskaya Cave. Confidence ellipses at 95%. Legend: A – Layer 1C, B – Layer 1B, C – Layer 1A-2, D – Layer 1A-1/1A-2, E – Layer 1A-1. There are probably no any other Chikiani obsidian sub-sources but 3a, and evidently it is the most commonly procured archaeological source, according to Biagi and Gratuze (2016) and Biagi et al. (2017).

brought to the cave from more distant sources located at linear distances of about 150–180 km NW. In the upper horizon (AH III), dating to LUP, from 29 to 24 ka cal BP, there is also evidence of sporadic contacts of Aghitu-3 inhabitants with more distant obsidian sources: Pokr Arteni and Meydan Dağ in eastern Turkey, located about 220 km and 250 km linearly from Aghitu-3 respectively.

The results of obsidian source analysis at Mezmaiskaya and the UP sites in the South Caucasus provide clear evidence that UP humans in the Caucasus transported obsidian over long distances and from various sources, which suggests large-scale movements of UP human groups across the landscape and exchange. The combined data on UP obsidian transport in three regions of the Caucasus—Mezmaiskaya in the NW

Caucasus, Aghitu-3 in the Lesser Caucasus, and the west Georgian sites of Bondi and Ortvale klde in the SW Caucasus—indicate that, taking into account that the access to obsidian sources did not involved crossing the high-mountain barrier of the Greater Caucasus (with modern mountain passes located at altitudes higher than 3500 m):

- obsidian was used as the prevailing raw material (only in Aghitu-3) when local or regional (according to Kandel et al., 2017; <50 km linearly from the site) obsidian sources were available;
- obsidian was used more or less regularly (~1–8% of total obsidian artifacts in Aghitu-3; ~1–2% of the total lithics in Dzudzuana, Ortvale Klde, and Bondi; ~0.5–3% of the total lithics in Mezmaiskaya)

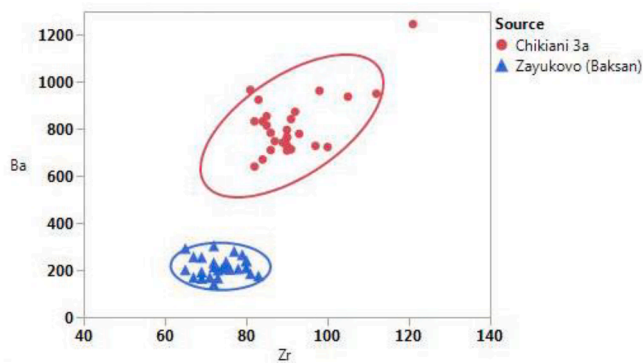


Fig. 7. Ba/Zr bivariate plot of the Mezmaiskaya obsidian artifacts using the scale of Biagi et al. (2017:14) and Biagi and Gratuze (2016) for the Chikiani obsidian source expanded to plot the Zayukovo produced artifacts. Allowing for inter-instrument variability, this suggests that all the Chikiani obsidian artifacts at Mezmaiskaya were produced from the Chikiani 3a chemical group as defined by Biagi et al. (2017), evidently a common procurement pattern. Confidence ellipses at 95%.

when it was derived from distant (100–250 km linearly from the site) obsidian sources.

- obsidian from long-distance (at linear distances of > 250 km from the site) sources was used only sporadically.

Bourdonnec Le et al. (2012:1328) proposed that in Bondi and Ortvale Klde, where obsidian was derived mainly from a single distant source (Chikiani), which involved a minimum walk of 170 km one way, “this might conceivably have been undertaken by the inhabitants themselves”, whereas for a few obsidian artifacts that probably entered the sites from long-distance sources (>200–250 km linearly; Ikisdere and Sarikamis in eastern Turkey, and Hatis and Gutansar in Armenia), with walking distances in excess of 350 km one way, they suggested “obsidian procurement likely involved some form of intermediary exchange.” However, later Pleurdeau et al. (2016:93) stated that the only source unambiguously identified in UP layers at Bondi and Ortvale Klde is Chikiani, and obsidian artifacts from the UP contexts in these sites only indicate “extensive long-range human mobility associated with this rock type.” Kandel et al. (2017: 64) didn’t specify whether the obsidian entered Aghitu-3 cave as a result of direct procurement via human mobility or exchange through long-distance social networks, while they concluded the long-distance (>200 km linearly) transport of obsidian identified in this cave should be included in behaviors associated with “an increase in the movements of individual groups across the landscape..., coupled with the creation and maintenance of more sophisticated social networks.”

The most long-distance obsidian transportation defined at Mezmaiskaya Cave (from the Chikiani obsidian source, ~450 km linearly SSE from Mezmaiskaya) is a unique case for the Caucasian UP, having no analogues in other UP sites in this region. Coming from the data discussed above, we propose the most relevant explanation for this phenomenon is that the Chikiani obsidian entered Mezmaiskaya Cave as a result of intermediary exchange through a long-distance social network established between related UP populations in NW and SW Caucasus. Future research of UP obsidian artifacts in Mezmaiskaya and increase of comparative data from other UP sites in the SW Caucasus are required to better understand how this obsidian transport network functioned during UP and how it was associated with other cultural exchange and social interactions between regional UP populations of the Caucasus.

4.2. Non-utilitarian artifacts as evidence of long-distance contacts via mobility

4.2.1. Azurite-quartz sandstone pebble

The Elbrus region in the NC Caucasus is notable as the area producing the only obsidian source (Zayukovo or Baksan; ~250 km linearly SE of Mezmaiskaya) known in the North Caucasus, which was used by UP humans at Mezmaiskaya Cave during the entire UP (see above). This strongly suggests that the UP population in the NW Caucasus had long-term contacts with the Elbrus region. The analysis of a pebble of azurite-quartz sandstone from Mezmaiskaya indicates that the rock, having such chemical composition and size, most likely originated from either Tyrnyauz (~250 km linearly SE of Mezmaiskaya) or Khudes (~200 km linearly SE of Mezmaiskaya) deposits located also in the Elbrus region and on the most plausible route, taking into account environmental parameters, from Mezmaiskaya Cave to the Zayukovo (Baksan) obsidian source area (Fig. 1). The pebble was brought to Mezmaiskaya most likely as a non-utilitarian object that UP humans picked up for its exceptional color and luster. This unusual find provides an additional confirmation of long-distance contacts via exchange or mobility between UP humans in the NW Caucasus and the Elbrus region in the NC Caucasus.

4.2.2. Black Sea marine mollusc shells

The marine gastropod shells found in the UP layers in Mezmaiskaya Cave is the only type of remains from marine organisms that was found so far in the site. All marine molluscs found at Mezmaiskaya are commonly known in the Black Sea basin. The nearest area where UP humans from Mezmaiskaya could collect these shells is the eastern coast of the Black Sea, located about 70 km linearly SW from Mezmaiskaya, but in modern relief separated from the cave by several high-mountain passes up to 1800 m asl through the Greater Caucasus Ridge (Fig. 1). The finds of marine shells suggest rare contacts of UP humans from Mezmaiskaya Cave with the Black Sea shore region, which is located on the most likely route of the obsidian transport/exchange network between the Chikiani source in SW Caucasus and NW Caucasus.

4.3. Evidence of contacts between UP populations of the NW Caucasus and Russian Plain

Among organic artifacts found in UP layers at Mezmaiskaya, objects made from mammoth tusk are of particular interest as possible evidence of long-distance contacts between the UP population of the NW Caucasus and some UP cultural entities that are known north of the Caucasus, on the Russian Plain. Besides Mezmaiskaya Cave, mammoth remains or artifacts made from mammoth tusk have not been found in any UP site in the Caucasus.

The flat stripe-beads made of mammoth tusk found in Layer 1A-2 (Fig. 2 – 2-4) have no analogues in the UP of the Caucasus. The closest analogy to these artifacts are stripe-beads made of mammoth tusk found in the Yudinovo site near Bryansk, on Russian Plain. Griborieva (2005: 33, 42) defined this type as ‘segment-like’ stripe-beads and noted that their shape and manufacturing technique are unique in comparison to stripe-beads that were found in many UP sites in East Europe and Siberia. The stripe-beads from Mezmaiskaya are similar in shape, size and manufacturing technique to the segment-like stripe-beads from Yudinovo, although Yudinovo is dated from the later UP, ~17–18 ka cal BP (Khlopachev and Gribchenko, 2012), much later than Layer 1A-2 (~27 ka cal BP) in Mezmaiskaya Cave.

The bead made from mammoth tusk and the pendant made of a tubular bone of a bird from Layer 1A-1/1A-2 in Mezmaiskaya (Fig. 2 – 6,7) also have no analogues in the UP of the Caucasus but are similar to artifacts made from bone and mammoth tusk that were found in the Sunghir site located on the Russian Plain (see Bader, 1978). Recent radiocarbon analyses date Sunghir site to the EUP, between 34.6 and 33.6 ka cal BP (Sikora et al., 2017; and references therein), older than the age of Layer 1A-1/1A-2 (~25 ka cal BP) at Mezmaiskaya. However,

Sunghir belongs to the well-known Streletskian industry which was widespread in East Europe during the EUP, between approximately 36–28 ka cal BP. Researchers proposed earlier that the Streletskian industry may represent a 'transitional' UP industry similar to the Central European Szeletian, and some Streletskian assemblages from Kostenki near Voronezh, such as Kostenki 1/Layer V and Kostenki 12/Layer III, can be dated from the earlier time (Anikovich et al., 2007). However, recent analyses suggest that these assemblages cannot be used to date the Streletskian industry, because they cannot be confidently associated with diagnostic Streletskian material (Dinnis et al., 2019). Also, the paleogenetic results from Sunghir indicate that the EUP population associated with the Streletskian industry exhibits genetic affinities with more western modern human groups associated with the Gravettian cultures in Europe (Sikora et al., 2017).

The southmost assemblage attributed to the Streletskian industry is known from the Biryuchiya Balka–2 open-air site located in the Seversky Donets River valley (the Don River lower basin), approximately 400 km north of Mezmaiskaya Cave (Fig. 1). In Biryuchiya Balka–2, the UP levels are dated to c. 31–26 ka BP (Matyukhin and Sapelko, 2010). Although bone artifacts were not preserved at Biryuchiya Balka–2, this site demarcates at present the southern most boundary of the Streletskian industry distribution.

Taking into consideration the palaeogenetic data from the Caucasus indicating no evidence of any genetic influx from European UP populations to the UP population of the Caucasus (Jones et al., 2015; Fu et al., 2016), cultural differences between the Streletskian industry and the LUP industry from Mezmaiskaya Cave and other Caucasian sites (see Golovanova and Doronichev, 2020), as well as the exceptional character of artifacts made from mammoth tusk found in Mezmaiskaya it is hard to assume any regular social relations via long-distance networks between the UP population in the NW Caucasus and the Streletskian or Gravettian populations on the Russian Plain. All artifacts from Mezmaiskaya having analogies in Sunghir are personal decorations that have some social and cultural meaning for hunter-gatherers groups. Apparently, their discovery at Mezmaiskaya suggests only that the UP humans of NW Caucasus could have some sporadic, irregular contacts with representatives of other UP cultural groups from Russian Plain. We think that future research, primarily direct palaeogenomic data for the UP population in the NW Caucasus, which are unknown at present, will provide more detail information to clarify the character of these contacts.

5. Conclusions

In conclusion, we suggest that the evidence from Mezmaiskaya Cave indicates that long-distance contacts between the UP humans in the NW Caucasus and neighboring regions had a different character and occurred for different reasons:

1. The rare presence in the cave of a high-quality lithic raw material (obsidian) from very distant sources had an utilitarian value. In addition, the transportation of obsidian to the NW Caucasus from the South Caucasus further confirms cultural similarity between the UP industries in these regions, as well as supports the hypothesis that UP groups spread northwards from southwestern Asia to both the South and North Caucasus (Golovanova and Doronichev, 2012). The data on obsidian artifacts from UP layers at Mezmaiskaya Cave suggest that there was a long-term and regular social networking between culturally related UP groups in the NW Caucasus and SW Caucasus during the entire UP, from c. 40–36 ka cal BP to the onset of LGM at c. 24–23 ka cal BP.
2. The rare Black Sea marine mollusk shells found in all UP layers at Mezmaiskaya, which could be used as personal decorations that had some social and cultural value, suggest sporadic contacts of UP humans in the NW Caucasus with the Black Sea shore region. Apparently, the presence of these exotic items in Mezmaiskaya Cave can be associated with the obsidian transportation/exchange

network to Mezmaiskaya from the Chikiani obsidian source in the South Caucasus, which route most likely followed along the Black Sea coast.

3. The unique finding of a pebble of azurite-quartz sandstone apparently should be seen as an exceptional example of an unusual, non-utilitarian object that UP humans picked up and brought to Mezmaiskaya Cave for its exceptional color and luster. Analyses of the azurite-quartz pebble suggest that the Tyrnyauz or Khudes deposits in the Elbrus region are the most likely sources of origin for this pebble. The appearance of this object in Mezmaiskaya Cave was likely associated with the obsidian transport network to Mezmaiskaya from the Zayukovo (Baksan) obsidian source, located also in the Elbrus region.
4. The personal decorations made from mammoth tusk found in Mezmaiskaya Cave, which have no analogies among other UP sites in the Caucasus but are similar to artifacts known from a very remote and culturally different region—the Streletskian UP site of Sunghir on the Russian Plain—can be assessed as the evidence of rare contacts between the UP population in the NW Caucasus and other culture-unrelated UP populations on the Russian Plain.

Finally, we conclude that the evidence for long-distance contacts, which we have recorded in Mezmaiskaya Cave in the NW Caucasus, delineates, at present, the borders of a cultural area occupied by related UP groups that established complex social relations via large-scale social and exchange networks. Also, the data from Mezmaiskaya suggests sporadic, occasional contacts of UP humans in the NW Caucasus with other UP cultural areas on Russian plain.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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